

Making Quantum Programming More Accessible with a Block-Based Language

Quantum computing is currently a major area of research which has the potential to solve problems which have been deemed impractical for conventional computers. However, programming quantum computers can be a daunting task for beginners and inexperienced programmers. While many quantum programming languages and frameworks already exist, such as Q# and Qiskit, these languages still require the user to have some familiarity with conventional programming concepts and/or mathematical notation in order to get any meaningful use out of them. This barrier to entry can make it hard for inexperienced programmers to experiment with quantum programming in order to learn the fundamentals of quantum computing.

This project will design and implement a simple block-based quantum programming language, inspired by educational programming environments such as Scratch. The aim of this project is to provide a more accessible way for inexperienced programmers to be introduced to quantum programming without requiring them to have preexisting knowledge of conventional programming languages. The product will allow users to construct quantum programs by combining visual blocks which will each represent a quantum operation such as applying a quantum gate or taking a quantum measurement. These programs can then be executed using a quantum simulator so that the user can learn by inspecting the result of their programs.

The project will also include a small case study at the end to investigate whether the block-based approach actually makes quantum programming more accessible for inexperienced programmers. Participants will be given programming tasks to complete using both the block-based language developed during the project and an existing quantum programming language or framework. Various metrics will be measured while the participants complete these tasks, such as time taken and number of errors, which will then be used to compare the accessibility of the block-based language versus the existing language.

Project goals:

1. Learn about quantum computing and quantum programming
2. Learn about visual/block-based programming language design
3. Design and implement a block-based quantum programming language
4. Conduct a small case study comparing the accessibility of the quantum block-based language versus an existing quantum language

Background:

<https://link.springer.com/article/10.1140/epjqt/s40507-025-00314-9>

QScratch is a similar project which introduces a block-based language used to teach quantum mechanics. While this project does include ideas from quantum mechanics, such as entanglement and measurement, it does not delve into the area of quantum computing and hence cannot represent or execute quantum circuits or algorithms.

The project described in this document instead aims to create a block-based language using a qubit and quantum gate model which is capable of running quantum algorithms such as Grover's algorithm.

This project will be supervised by Jeremy Singer and Jacob Trevor.